

LUMINOUS CAMPUS

PRODUCTION SPECIFICATION

Platform Architecture, Security System & Academic ERP Overview

Target: Next.js 16 + React 19 + Supabase + Gemini
Date: August 2026

Executive Summary: Luminous is an enterprise campus operating system uniting **high-stakes safety operations** (2-tier SOS, real-time CAD vector map, SLA timers, AI incident triage) with **complete academic ERP governance** (21 modules spanning courses, attendance, exams, hostels, placements, and wellbeing).

1 Frontend & Backend Technology Stack

FRONTEND TECHNOLOGY STACK	BACKEND & DATA INFRASTRUCTURE
Framework: Next.js 16.3.1 (App Router, Turbopack, SSR/SSG) UI Engine: React 19.2.8 & TypeScript 5 (<code>strict: true</code>) Styling: Tailwind CSS v4.3.3 (CSS-first <code>@import</code> design tokens) Component Layer: Native React 19 primitives (zero Radix bloat) Visualizations: Recharts 3.10 + Custom SVG CAD Map Iconography: Lucide React (800+ consistent icons) State: 4 Domain Contexts (Auth, Academic, Safety, Services)	API Routing: Next.js Route Handlers + Edge Proxy (<code>proxy.ts</code>) Database & Auth: Supabase (PostgreSQL with RLS & JWTs) SSR Session Auth: <code>@supabase/ssr</code> (Cookie session synchronization) AI Intelligence: Google Gemini 2.0/3.7 Flash + Heuristics Fallback Validation: Zod v4.4 schemas on all routes and AI outputs Security Engine: In-memory sliding rate limiters, CSRF, crypto IDs

2 Stack Justification: Why These vs. Alternatives

Technology	Alternatives Rejected	Core Rationale & Advantage
Next.js 16	Vite, CRA, Remix, Nuxt	Unifies SSR, Edge Proxy, Route Handlers, and React 19 in one project. CRA is deprecated; Vite lacks native Edge proxy middleware.
Supabase (PostgreSQL)	Firebase (Firestore), AWS Amplify	Relational schema is mandatory for ERP joins (Student - Course - Attendance - Incident). Firebase NoSQL cannot handle multi-table academic queries.
Gemini 2.0/3.7 Flash	OpenAI GPT-4o, Claude 3.5	Sub-200ms emergency triage latency at ~1/20th token cost. Paired with deterministic rule fallback for zero-downtime safety compliance.
Tailwind CSS v4	Bootstrap 5, MUI, Emotion	Zero runtime CSS overhead (~10KB purged bundle). Unlocks complete custom branding (Luminous Gold & Frosted Silver) without framework overrides.
Native Primitives	Radix UI, Ant Design, Chakra	Custom React 19 primitives save ~200KB bundle weight while maintaining full accessibility and eliminating external library vulnerabilities.

3 Security & Emergency Safety System (In-Depth)

High-Stakes Campus Safety: The safety architecture guarantees immediate emergency distress reach, automated SLA escalation, AI-assisted triage, and zero-trust data protection.

Seven-Layer Defense-in-Depth Security Pipeline

PERIMETER & NETWORK DEFENSE

- **Layer 1 (Edge Headers):** CSP, HSTS (``max-age=2yr``), X-Frame SAMEORIGIN, Permissions-Policy (GPS allowed, Mic blocked).
- **Layer 2 (CSRF Guard):** Rejects cross-origin fetch (``sec-fetch-site``) & validates Origin/Referer.
- **Layer 3 (Rate Limiter):** Sliding window limits: SOS (10/min), Alerts (5/min), Copilot (25/min), Auth (20/min).

IDENTITY, DATA & PAYLOAD SECURITY

- **Layer 4 (Auth Guard):** Supabase Bearer JWT & cookie session hydration.
- **Layer 5 (RBAC Gateway):** 9 roles; fail-secure deny-by-default routing.
- **Layer 6 (Zod Schemas):** Validates payloads and regex-sanitizes URL inputs.
- **Layer 7 (Crypto IDs):** ``crypto.randomUUID()`` non-enumerable resource IDs.

Emergency Response & Safety Features

Safety Feature	Operational Mechanism	User Benefit & Emergency Impact
Two-Tier SOS Beacon <code>/safety/sos</code>	Level 1 (Campus): 3 taps dispatches nearest guard + starts 5-min SLA timer. Level 2 (Police): 5 taps/hold escalates to multi-squad and police.	Guaranteed instant response; automatic escalation if guards do not resolve within 5 minutes. Live GPS broadcast.
Emergency Broadcasts <code>/safety/emergency</code>	Role-gated dispatches scoped by: <i>Campus-wide, Building, Hostel Block, or Department.</i>	Eliminates mass panic by targeting only affected zones during local fires, lab leaks, or weather hazards.
AI Incident Triage <code>/api/ai/classify</code>	Gemini Flash classifies severity (LOW to CRITICAL), summary, and assigns response unit with deterministic fallback.	Lab smoke/hazard automatically triggers immediate electrical isolation and Level 1 evacuation directives in milliseconds.
Command Center & CAD Map <code>/campus-map</code>	Interactive SVG architectural vector blueprint with pulsing hazard pins, patrol units, and facility inspection.	Gives security officers single-pane spatial awareness of active campus hazards and nearest available guards.
FERPA-Compliant Copilot <code>/copilot</code>	Zero-trust tool authorization. Students see only own data; Parents see only linked child; Whistleblower IDs redacted.	Natural language assistance without exposing confidential student grades, records, or reporter identities.

4 ERP Platform Architecture & Institutional Benefits

Luminous is a unified **21-Module Higher Education ERP** that eliminates disconnected point solutions.

ACADEMIC GOVERNANCE (7 MODULES)

- **Departments:** Budget allocations, lab counts, HOD governance.
- **Courses & Syllabi:** Credits, semesters, instructor allocations.
- **Faculty Management:** Profiles, cabins, research, teaching hub.
- **Classroom Attendance:** Batch marking, defaulter alerts (<75%).
- **Examinations:** Mid-Term/Final scheduling, SGPA analytics.
- **Master Timetable:** Grid/Agenda views with room clash prevention.

STUDENT & CAMPUS OPERATIONS (7 MODULES)

- **Student Directory & Portal:** Roster, CGPA, daily schedule.
- **Parent Cockpit:** Ward attendance, grades, curfew visibility.
- **Hostel & Curfew:** Room allocations, maintenance, night logs.
- **Placement Services:** Corporate drives, auto-CGPA eligibility.
- **AI Grievance Redressal:** 8-domain auto-ticket routing.
- **Student Wellbeing:** Confidential counseling & crisis lines.

The ERP Cross-Module Advantage (Zero Data Silos)

- **Exams to Placements:** Published CGPA and backlogs in `/exams` automatically lock/unlock eligibility for company drives in `/placement`.
- **Attendance to Parent Portal:** Faculty submissions in `/attendance` immediately reflect in `/parent` for attendance defaulter warnings.
- **Hostel to Security:** Curfew turnstile anomalies in `/hostel` immediately trigger patrol alerts in `/security`.
- **Safety to Interactive Map:** Reports filed in `/incidents` project live hazard beacons onto the vector CAD blueprint in `/campus-map`.

5 Future Expansion Roadmap: What Else Can We Do?

NEAR-TERM (PHASE 1)

- **Web Push / PWA:** Emergency push alerts even when browser is closed.
- **Dark Mode UI:** Night-shift theme for security desk operations.
- **PDF/CSV Export:** Attendance sheets, grade cards, audit logs.
- **SMS/Email Fallback:** Twilio SMS for critical campus emergencies.

MEDIUM-TERM (PHASE 2)

- **Mobile Native Apps:** React Native / Flutter apps with offline SOS.
- **CCTV Camera Stream:** Real-time feeds in Command Center with AI anomaly alerts.
- **Tuition & Fee Gateway:** Payment reconciliation and scholarship tracking.
- **Fleet GPS:** Live campus shuttle tracking and QR transport passes.

LONG-TERM VISION (PHASE 3)

- **Smart Building IoT:** Smoke/gas detectors feeding directly into Safety Context.
- **Multi-Campus Federation:** Centralized governance across university branches.
- **AI Vision Surveillance:** Automated perimeter breach & crowd analytics.
- **Voice-Activated SOS:** Hands-free voice emergency trigger ("Hey Luminous").